

Dermatology Assistant

Project Details:

Project's main aim is to gather data from patient like his lesion image, his medical conditions, details related to skin disease like how much time before it occurred, has the skin rashes increased lately, is there an itch to scratch the lesion part/ skin increased lately, is skin lesion area increased on skin. Now this project needs to be done so that skin lesion cancer classification is done easily. Skin lesion images are done using a camera (phone/laptop). Now, if a patient can't explain his conditions, he can give his skin biopsy report and blood test report (can check if elevated levels of lactate dehydrogenase (LDH)) to the bot and after getting every information, process it in models and get an accurate diagnosis of patient's condition. The whole process described above must happen while the bot chats continuously with the patient and chat history must be saved for the same process. Now after all this understanding is done, let's move forward to defining the stages of project.

Stages:

1. Preprocessing Datasets for inputs to model to be trained
2. Skin Lesion Segmentation (SLS):

Now our input skin lesion is a large image which contains both skin lesions and a large part of plain skin with hair/no hair and color. The first task is to convert a large skin image into b/w only lesion image using segmentation. This

process helps us get a scanned image of lesions and remove extra skin parts.

3. Skin Lesion ROI Cropping(SLRC)

From above stage, after obtaining the B/W segmented skin lesion image, now this stage helps to get the original color image of lesion by combining the original skin lesion image with b/w segmented lesion image. This may help the segmented image to convert from b/w to its original color making sure that it covers all the borders of the skin lesion image required. SLS and SLRC stages help the input image ready for classification and analysis process.

4. Hierarchical Classification + ABCDE values

For this stage, input for classification is taken from the SLRC stage. It helps to classify the skin lesion as cancerous and non cancerous first and then helps the lesion classification in their sub categories respectively. For ABCDE values, A (Asymmetry) is checked while performing SLS stage, B (Border Irregularity) is checked if it is regular or irregular, C (Color Variation) is to check color attained in lesion part. D (Diameter) is checked when doing the SLS stage only by assessing pixel ratio. E (Evolution) can be done by asking patient questions like has your lesion size increased or any other questions or if by luck patient has the image of lesion taken some days prior. This stage handles the most task of analysing lesions and its problems. This stage returns the classification and ABCDE values need to be stored for analysis. It also needs to handle risk value for each type of skin disease related to lesion

5. MedProc:

This stage can help to analyse the skin lesions ABCDE values using the calculated ones at HC stage and then process the information on patients condition which can be related or non related to skin disease. Using the model knowledge and its capability, ABCDE values can be used to define a patient's lesion condition as well as the possibility of melanoma or non melanoma using ABCDE details. This can follow a rule based model since ABCDE values are following mostly some rules. Now other non related diseases can be handled by hospital dataset so it follows normal classification + regression based model based on disease and descriptions by patient. As this bot acts as some general physician, it may suggest patient some medicine (eg. paracetamol for cold, pantoprazole for acidity, domperidone for indigestion, etc). This gives the output in JSON format for reasoning of every problem.

6. Image+Text Fusion Model (IT Fusion):

Now this layer is a fusion of MedProc and HC stage where classification process of HC and medical data analysis of given skin image + extra non related diseases is to be handled. This model needs to handle every input and output meticulously and give the result regarding classification data, risk, ABCDE values, its analysis and any other meaningful / related things to this process. In this stage, we need to handle preprocessing, SLS, SLRC so that HC and MedProc get their cleaned inputs. This stage output needed to be stored in JSON so while VQA we can get all the responses to the question in easy format

7. Virtual Question Answering (VQA):

This stage is mainly responsible for analysing the data gathered from the IT stage model and using them to answer the questions asked by the bot respectively. Bot is given a question from the patient which is then given to the VQA model which then turns to gather data from the patient and using some rules VQA model is to be made sure at what time it needs to send which data to which model. This helps the VQA understand what data needs to be sent to which model and for answering part VQA needs to understand from which part of JSON content he needs an answer which is matching to question context and it should be accurately correct. Using Reinforcement Learning here to help the model pick the correct answer and using the algorithm's reward system, it needs to handle bot responses to questions in the given context making sure bot answers to question correctly with reasoning and without hallucination.

8. Frontend-Backend and API

This stage helps to make an Application Interface on a web browser using MERN stack and using Fast API which is connecting functions and models in python to show answers to patients in a representable and understandable way.

9. Testing and Deployment

Testing Stage can be initialized once all above stages are done and dusted. This stage is mostly done to check if the accuracy of models in the project can be increased and also to check data compatibility and data integration

required or not. In this stage whole project needed to be tested so the application made is sustainable and optimal. Once tested successfully, this application can be deployed

Dataset

Stage Name	Dataset Name(s)	Trainin g	Testin g	Validatio n	Total
SLS (Skin Lesion Segmentation)	ISIC 2016 – Task 1 Segmentation	900	379	100	1379
SLS – Extended	ISIC 2017 – Task 1 Segmentation	2000	600	300	2900
SLRC (ROI Cropping)	ROI crops generated from ISIC 2016 + ISIC 2017 segmentation outputs	2900	979	400	4279
HC (Level 1 – Benign vs Malignant)	ISIC 2018 Classification Dataset (10,015 images)	7000	2000	1015	10,015
HC – Combined ISIC Stages	ISIC 2019 (25,331 images) + ISIC 2020 (33,126 images)	40,000	10,000	8,457	58,457
HC – Extended Multi-Class Dermatology	ISIC 2016 Task 3 + ISIC 2016 Task 3B + ISIC 2017 Classification + ISIC 2024 (~18k images)	20,000	8,000	2,900	30,900
HC – Large Clinical Dataset	BCN-20000 (20,000 clinical dermatology images)	14,000	4,000	2,000	20,000
HC – Dermoscopic Pigmented Lesions	MiLK-10K (10,015 dermoscopic images)	7000	2000	1015	10,015

MedProc (Medical Text Understanding)	MIMIC-III + MIMIC-IV Clinical Notes	50,000	10,000	5,000	65,000
IT Fusion (Image + Text Fusion Model)	Derm1M (1M images) + HAM10000 + PAD-UFES-20 + ISIC 2016-2024 + BCN-20000 + MIMIC-III/IV clinical text	700,000	200,000	100,000	1,000,000 +
VQA (Virtual Question Answering)	DermaVQA + SkinCancerVQA + SkinQA (clinical) + PathVQA + ISIC-Derived DermaQA	80,000	15,000	10,000	105,000

Models

Stage name	Model name	Paper name	Is code available	Paper link	Paper citation	Use of model	Advantage of model	Disadvantage of model
SLS (Segmentation)	ARCUNet	ARCUNet: Enhancing Skin Lesion Segmentation with Residual Connections and Attention	Yes (NIH/PMC article typically links to code)	https://pubmed.ncbi.nlm.nih.gov/articles/PMC11920274/		Dermatology-focused U-Net variant with residual and attention modules for skin lesion segmentation.	Designed specifically for skin lesions; strong boundary accuracy and easy to integrate into existing U-Net pipelines.	Not necessarily the absolute top performer compared to newest transformer hybrids; slightly heavier than plain U-Net.

HC (Classification)	EfficientFormerV2	EfficientFormerV2: A Lightweight Vision Transformer for Efficient Image Recognition	Yes (official GitHub)	https://github.com/snnap-research/EfficientFormer		Backbone for benign vs malignant and multi-class lesion classification on dermoscopy/clinical crops.	Very efficient transformer backbone with good accuracy and low VRAM usage; suitable for multi-head HC on 12 GB GPU.	Not dermatology-specific; may slightly trail heavier Swin/ViT backbones on peak accuracy if you can afford them.
MedProc (medical text)	BioBERT	BioBERT: A Pre-trained Biomedical Language Representation Model for Biomedical Text Mining	Yes (Hugging Face, etc.)	https://spacknlp.org/2020/08/25/scient-biobert-clinical_base_cased.html		Text encoder for biopsy reports, blood test summaries, and medical descriptions.	Strong performance on biomedical NER and classification; widely used and supported.	Optimized mainly for literature-style biomedical text; not as strong on long, messy EHR notes as newer clinical models.
IT Fusion (handling missing modalities)	MARIA-style transformer	A Multimodal Transformer Model for Incomplete Healthcare Data (MARIA)	Likely yes (arXiv)	https://arxiv.org/abs/2412.14810		Fusion transformer that combines multiple modalities (image, labs, text, structured data) via masked attention to handle missing modalities.	Naturally handles incomplete modality sets, matching your real-world scenario where not all patients have all tests.	Requires custom integration of your encoders; not dermatology-specific; more engineering effort than simple concatenation.

VQA (baseline VLM)	R-LLaVA	R-LLaVA: Improving Med-VQA Understan ding Through Visual Region of Interest Guidance	Yes (OpenRev iew/GitHu b)	https://ope nreview.net /forum?id= HqdlX8yZJ P		Med-VQA model that uses ROI informatio n to guide visual attention and improve answers.	Aligns perfectly with your SLRC stage by using lesion ROIs; outperfor ms standard LLaVA-Me d on multiple Med-VQA benchmar ks.	General Med-VQA, not dermatolog y-specific; needs domain-sp ecific fine-tuning and potentially model-size tuning for 12 GB GPU.
SLRC (Skin Lesion ROI Cropping)	OpenCV							
Frontend- Backend and API	React - Frontend , FastAPI- linking							

Literature Survey

Author Name	Year	Paper Title	Link	Difference from NutriDermAI	Insights Gained	Cit atio ns
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Pan et al.	2025	A Multimodal Vision Foundation Model for Clinical Dermatology (PanDerm)	https://www.nature.com/articles/s41591-025-03747-y	Large dermatology foundation model across multiple imaging modalities; does not implement full conversational assistant, ABCDE workflow, or RL-based VQA like NutriDermAI.	Shows how self-supervised masked latent modeling and CLIP-style alignment on millions of dermatology images yields universal encoders reusable for SLS, HC, and IT Fusion.	Pan et al. (2025). A Multimodal Vision Foundation Model for Clinical Dermatology (PanDerm) [https://www.nat
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Zeng et al.	2 0 2 5	MM-Skin: Enhancing Dermatology Vision-Language Model with Expert-Curated Multimodal Dataset	https://arxiv.org/ html/2505.0615 2v1	Builds dermatology VLM on expert-captioned multimodal data; lacks explicit MedProc rule engine, ABCDE metrics, and end-to-end chat deployment.	Demonstrates that expert-written long-form descriptions and multi-modality inputs significantly improve image–text understanding, guiding IT Fusion training and prompting strategies. Zen g et al. (20 25) . M M- Ski n: Enh anc ing Der ma tolo gy Visi on- Lan gua ge Mo del wit h Exp ert-

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Yan et al.	2025	Derm1M: A Million-Scale Vision-Language Dataset Aligned with Hierarchical Dermatology Ontology	https://arxiv.org/abs/2503.14911	Provides 1M+ image-text pairs focused on pretraining; does not implement a staged assistant with SLS, ABCDE, and RL-based VQA.	Enables robust dermatology vision-language pretraining where DermLIP outperforms biomedical foundations in zero-shot classification, strongly informing IT Fusion backbones. Yan et al. (2025). Derm1M: A Million-Scale Vision-Language Dataset Aligned

						with Hierarchic al Dermatology Ontology [https://arxiv.org/abs/2503.14911]
Aboulmira et al.	2025	Attention-Guided Multimodal Skin Disease Classification	https://www.sciencedirect.com/science/article/pii/S2352914825000887	Combines dermoscopic images and text in an attention-guided classifier; does not include conversational VQA, MedProc reasoning, or multi-stage ISIC/BCN integration.	Shows how attention-based fusion of image and text can improve diagnosis, suggesting architectural patterns for NutriDermAI's multimodal IT Fusion stage.	Aboulmira et al. (2025). Attention-Guided Multimodal Skin Disease

						Classification [https://www.sciencedirect.com/science/article/pii/S235291482500887]
Naseem et al.	2024	BioMedBLIP: Advancing Accuracy in Multimodal Medical Tasks Through Biomedically-Informed BLIP Models	https://medinform.jmir.org/2024/1/e56627	Targets radiology and other modalities; not specialized for dermatology or ABCDE-driven workflows.	Demonstrates that domain-adapted BLIP models outperform generic ones on medical tasks, informing model choice and pretraining strategy for IT Fusion and VQA.	Naseem et al. (2024) . BioMedBLIP: Advancing Accuracy in Mul

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ScaleFusionNet authors	2025	Transformer-Guided Multi-Scale Feature Fusion for Skin Lesion Segmentation	https://arxiv.org/html/2503.03327v3	Focuses on high-accuracy segmentation using Swin Transformers and deformable convolutions; does not implement ROI	Shows multi-scale fusion with Transformers significantly boosts Dice scores on ISIC, making such architectures	Sca leF usi on Net aut hor s (20

			color reconstruction, ABCDE scoring, or multimodal reasoning.	strong SLS candidates feeding SLRC and downstream modules.	25) . Transformer-Guided Multi-Scale Feature Fusion for Skin Lesion Segmentation [https://arxiv.org/html/2503.03327v3]
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ESC-UNet authors	2025	ESC-UNET: A Hybrid CNN and Swin Transformers for Skin Lesion Segmentation	https://www.sciencedirect.com/science/article/pii/S2666521225000614	Hybrid CNN–Swin model restricted to segmentation; not embedded in a full conversational assistant or VQA pipeline.	Confirms that combining CNN backbones with Swin Transformers yields high performance on ISIC data, supporting similar hybrid choices for NutriDermAI’s SLS.	ESC-UNet authors (2025). ESC-UNET: A Hybrid CNN and Swin Transformers for Skin Lesion Segmentation [https://www.sciencedirect.com/sci
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MRP-UNet authors	2 0 2 5	Skin Lesion Segmentation with a Multiscale Input Fusion U-Net	https://www.nature.com/articles/s41598-025-92447-1	Improves segmentation boundaries via multiscale input fusion; does not support ROI reconstruction or text fusion as required in SLRC and IT Fusion.	Demonstrates that multiscale fusion and residual attention refine lesion borders and small-lesion sensitivity, directly benefiting SLS and ABCDE A/B estimation.	MR P-U Net aut hor s (20 25) . Ski n Les ion Seg me nta tion wit h a Mul tisc ale Inp ut Fus ion U-N et [htt ps: //w ww. nat ure. co

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Meta-UNet authors	2 0 2 5	Meta-UNet: Enhancing Skin-Lesion Segmentation with Multimodal Metadata	https://pmc.ncbi.nlm.nih.gov/articles/PMC12476315/	Uses metadata to refine segmentation masks; not connected to VQA, ABCDE rule logic, or conversational pipeline.	Shows that including metadata such as lesion location and patient info improves segmentation, encouraging early fusion of structured data in SLS and IT Fusion.
					Me ta- UN et aut hor s (20 25) . Me ta- UN et: Enh anc ing Ski n-L esi on Seg me nta tion wit h Mul tim oda l Me tad ata [htt ps:

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BiADATU-Net authors	2024	Intelligent Skin Lesion Segmentation Using Deformable Attention Transformer U-Net	https://onlinelibrary.wiley.com/doi/10.1111/srt.13783	Focuses purely on deformable-attention-based segmentation; does not integrate text or MedProc reasoning.	Highlights that deformable attention captures irregular lesion borders, improving boundary precision and supporting robust ABCDE border feature extraction.	BiADATU-Net authors (2024). Intelligent Skin Lesion Segmentation Using Defor- ma- ble Att- enti- on Tra

						nsf or me r U-N et [htt ps: //o nlin elib rary .wil ey. co m/ doi /10 .11 11/ srt. 137 83]
MUCM-Net authors	2024	MUCM-Net: A Mamba Powered UCM-Net for Skin Lesion Segmentation	https://www.explorationpub.com/Journals/em/Article/1001250	Provides lightweight segmentation; does not embed in a larger multimodal, conversational framework.	Delivers an efficient segmentation backbone suitable for 12GB GPUs, aligning with NutriDermAI's hardware constraints for the SLS stage.	MU CM -Ne t aut hor s (20 24) · MU CM -Ne t: A Ma mba Po wer ed UC M- Net

					for Ski n Les ion Seg me nta tion [htt ps: //w ww. exp lora tion pub .co m/ Jou rnal s/e m/ Arti cle/ 100 125 0]
Hybrid dual-encoder authors	2 0 2 4	A Hybrid Deep Learning Approach for Skin Lesion Segmentation Using Dual CNN-ViT Encoders	https://ieeexplore. iee. org/iel8/6 287639/108201 23/10910106.pd f	Uses ResNet + ViT dual encoders for segmentation only, without downstream ROI reconstruction or multimodal heads.	Shows that dual-encoder CNN-ViT architectures with attention fusion improve segmentation, offering reusable patterns for SLS and feature extraction feeding IT Fusion. Hy brid dua l-en cod er aut hor s (20 24) . A Hy brid Dee p Lea rnin

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Faggioli et al.	2025	Multimodal Learning for Skin Lesion Segmentation and Closed-Ended Visual Question Answering	https://www.dei.unipd.it/~faggioli/temp/clef2025/paper_197.pdf	Jointly tackles segmentation and closed-ended VQA for MEDIQA-MAGIC; does not implement full HC, MedProc, or dialogue management.	Confirms that shared representations between SLS and VQA benefit performance, supporting tight integration of SLS/SLRC with VQA in NutriDermAI.	Faggioli et al. (2025). Multimodal Learning for Skin Lesion Segmentation and Closed-Ended Visual Question Answering [https://www.dei.unipd.it/~faggioli
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IReL team	2025	Tackling Multimodal Dermatology with CLIPSeg-Based Vision-Language Fusion	https://ceur-ws.org/Vol-4038/paper_208.pdf	Uses CLIPSeg-based segmentation and multimodal alignment for challenge tasks; lacks ABCDE scoring, rule-based MedProc, and RL answer selection.	Shows CLIP-style joint image-text representations help dermatology segmentation and VQA, informing CLIP-like encoders for IT Fusion and SLS.	IReL team (2025). Tackling Multimodal Dermatology with CLIPSeg-Based Vision-Language Fusion [https://ceur-

						ws. org /Vo l-40 38/ pap er_ 208 .pd f]
Review authors	2 0 2 4	A Review of U-Net-Based Deep Learning Models for Skin Lesion Segmentation	https://onlinelibrary.wiley.com/doi/10.1111/ima.70107	Survey of U-Net variants; does not implement a new pipeline or multimodal assistant.	Summarizes segmentation architectures and performance on standard datasets, guiding initial SLS backbone and baseline selection.	Rev iew aut hor s (20 24) . A Rev iew of U-N et- Bas ed Dee p Lea rnin g Mo del s for Ski n Les ion Seg me nta tion [htt ps: //o nlin

					elibrary. com/doi/10.1111/ima.70107]
Multimodal Skin Cancer Prediction authors	2025	Multimodal Skin Cancer Prediction: Integrating Dermoscopic Images and Multi-Modal Data	https://openbioinformaticsjournal.com/VOLUME/18/ELOCATOR/e18750362358444/	Integrates multiple data modalities for prediction but does not extend to conversational interaction, ABCDE rule logic, or VQA.	Shows benefits of fusing dermoscopic images with clinical data, supporting NutriDermAI's plan to combine HC outputs with MedProc in IT Fusion. Multimodal Skin Cancer Prediction: Integrating

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Jütte et al.	2024	Integrating Generative AI with ABCDE Rule Analysis for Early Skin Cancer Detection	https://www.frontiersin.org/journals/medicine/articles/10.3389/med.2024.1445318/full	Uses generative AI plus ABCDE analysis for detection without building a multi-stage assistant with VQA, MedProc, and deployment stack.	Provides a working example of ABCDE-aware generative models for skin cancer, validating NutriDermAI's ABCDE-centric risk analysis design.	Jütte et al. (2024). Integrating Generative AI with ABCDE Rule Analysis for Early Skin Cancer Detection [https://www.frontiersin.org/journals/
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Chatterjee et al.	2021	Dermatological Expert System Implementing the ABCD Rule for Melanoma Detection	https://www.sciencedirect.com/science/article/abs/pii/S0957417420309337	Implements rule-based ABCD expert system using classical processing, not deep segmentation, multimodal fusion, or conversational flow.	Offers formalization of ABCD criteria and total dermoscopic scores as rules, directly informing MedProc's rule-based component.	Ch att erje e et al. (20 21) . Der ma tolo gic al Exp ert Sys tem Imp lem enti ng the AB CD Rul e for Mel ano

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Senan et al.	2 0 2 1	Analysis of Dermoscopy Images by Using ABCD Rule for Melanoma Diagnosis	https://www.sci encedirect.com/ science/article/ pii/S2666285X2 1000017	Applies ABCD rule with classical methods; no VQA, LLMs, or multi-stage integration.	Reinforces clinical validity of ABCD rule and shows practical algorithms for automatic ABCD evaluation feeding HC and MedProc. Sen an et al. (20 21) · An aly sis of Der mo sco py Ima ges by Usi

					ng AB CD Rul e for Mel ano ma Dia gno sis [htt ps: //w ww. sci enc edir ect. co m/ sci enc e/a rticl e/p ii/S 266 628 5X 210 000 17]
Tas et al.	2 0 2 2	Faster Evaluation of ABCD Rule and Total Dermoscopic Score for Nevomelanocyt ic Lesions	https://brieflands.com/journals/ijssc/articles/102138	Optimizes ABCD/TDS computation speed; does not integrate into a multi-module conversational pipeline.	Demonstrates that ABCD/TDS evaluation can be accelerated for near real-time use, supporting responsive ABCDE scoring in interactive sessions. Tas et al. (20 22) · Fas ter Eva lua tion of

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Shetty et al.	2022	Skin Lesion Classification of Dermoscopic Images Using Hybrid CNN Models and ABCD Rule Features	https://www.nature.com/articles/s41598-022-22644-9	Combines CNN features with ABCD descriptors for classification but omits text reasoning, MedProc, and conversational frontends.	Validates the benefit of explicitly incorporating ABCD-based features into deep classifiers, informing HC feature engineering.	Shetty et al. (2022). Skin Lesion Classification of Dermoscopic Images Using Hybrid CNN Models and ABCD Rule Features [https://www.nature.co
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Mevorach et al.	2025	A Comparison of Skin Lesions' Diagnoses Between AI-Based Systems and Dermatologists Using ABCDE Rule	https://pmc.ncbi.nlm.nih.gov/articles/PMC12071753/	Evaluates existing AI systems vs dermatologists on ABCDE-based diagnosis; does not propose a new architecture.	Provides clinical evidence that ABCDE-based AI assessment can approach dermatologist performance, supporting NutriDermAI's ABCDE emphasis.

					Using ABCDE Rule [https://pmc.ncbi.nlm.nih.gov/articles/PMC12071753/]
Yim et al.	2024	DermaVQA: A Multilingual Visual Question Answering Dataset for Dermatology	https://dl.acm.org/doi/10.1007/978-3-031-72086-4_20	Releases dermatology VQA dataset and baselines but not a full multi-stage assistant with ABCDE, MedProc, and RL-based VQA.	Defines dermatology-specific VQA tasks and multilingual questions, guiding VQA training targets and evaluation for NutriDermAI.

						ing Dat ase t for Der ma tolo gy [htt ps: //dl .ac m.o rg/ doi /10 .10 07/ 978 -3-0 31- 720 86- 4_2 0]
Saeed et al.	2024	MediFact at MEDIQA-M3G 2024: Medical Question Answering with Weakly Supervised Answer Generation	https://aclanthology.org/2024.clinalnlp-1.31/	Uses weak supervision for answer generation in multimodal medical tasks; lacks explicit ABCDE computation and full lesion pipeline.	Shows how weakly supervised and knowledge-augmented training can compensate for scarce labels in medical VQA, applicable to NutriDermAI's training regime.	Saeed et al. (2024). MediFact at MEDIQA-M3G 2024: Medical

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Kim et al.	2024	TEAM MIPAL at MEDIQA-M3G 2024: Large VQA Models for Multimodal Medical Answer Generation	https://aclanthology.org/2024.clinalnlp-1.30.pdf	Adapts large VQA models for dermatology tasks but does not decompose pipeline into SLS, HC, MedProc, or use RL answer selection.	Provides fine-tuning and answer ranking strategies for large multimodal VQA, informing prompt and scoring design for NutriDermAI's VQA stage.	Kim et al. (2024). TEAM MIPAL at MEDIQA-M3G 2024: Large VQA Models for Multimodal Medical Answer Generation [https://aclanthology.org/2024.clinalnlp-1.30.pdf]
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H3N1 Lab	2025	DermKEM: A Knowledge-Enhanced Ensemble for Dermatology VQA	https://eur-ws.org/Vol-4038/paper_202.pdf	Ensembles VLMs with external knowledge; does not tie in ABCDE scoring or structured MedProc outputs as knowledge sources.	Shows that knowledge-enhanced ensembles improve VQA accuracy, suggesting integration of MedProc's JSON reasoning into VQA as structured knowledge.	H3 N1 Lab (20 25) . Der mK EM: A Kn owl edg e-E nha nce d Ens em ble for Der ma tolo gy VQ A [htt ps: //c eur- ws. org /Vo l-40 38/

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Madavan et al.	2025	Med-GRIM: Enhanced Zero-Shot Medical VQA Using Prompt Ensembles	https://openaccess.thecvf.com/content/ICCV2025W/MRR%202025/papers/Madavan_Med-GRIM_Enhanced_Zero-Shot_Medical_VQA_using_prompt-e.pdf	Targets zero-shot Med-VQA using prompt ensembles; not dermatology-specific and lacks RL-based training.	Demonstrates that combining prompts and VLMs improves zero-shot performance, inspiring ensemble and reranking strategies for NutriDermAI's VQA.	Madavan et al. (2025). Med-GRIM: Enhanced Zero-Shot Medical VQA Using Prompt Ensembles [https://openaccess.thecvf.co

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Li et al.	2022	Medical Visual Question Answering via Corresponding Feature Fusion and Semantic Attention	https://www.aimspress.com/article/doi/10.3934/mbe.2022478	Designs semantic attention and feature fusion for Med-VQA on radiology, not dermatology.	Introduces corresponding feature fusion between image and question with semantic attention, useful for designing NutriDermAI's IT Fusion/VQA heads.	Li et al. (2022). Medical Visual Question Answering via Corresponding Feature Fusion and Semantic Attention [https://www.aimspress.com/article/
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						doi /10 .39 34/ mb e.2 022 478]
DSAP authors	2024	Bridging the Semantic Gap in Medical Visual Question Answering with Dynamic Semantic-Adaptive Prompting	https://ieeexplore.ieee.org/iel8/42/11230061/11039185.pdf	Uses adaptive prompting to improve Med-VQA; not tailored to dermatology or ABCDE.	Shows dynamic prompts can narrow the semantic gap between image and question, suggesting a prompt-adaptor layer for NutriDermAI's VQA.	DSAP authors (2024). Bridging the Semantic Gap in Medical Visual Question Answering with Dynamic Semantic-Ad

					https://ieeexplore.ieee.org/iel8/42/1230061/11039185.pdf
MedVLM-R1 authors	2025 02 25	MedVLM-R1: Incentivizing Medical Reasoning Capability with GRPO for VQA	https://papers.miccai.org/miccai-2025/paper/3267_paper.pdf	Applies GRPO RL to radiology VQA; not specifically for dermatology or ABCDE scoring.	Provides an RL framework with format and accuracy rewards for VQA, directly informing RL-based training for dermatology VQA in NutriDermAI. MedVLM-R1 authors (2025). MedVLM-R1: Incentivizing Medical Reasoning

						ing Ca pab ility wit h GR PO for VQ A [htt ps: //p ape rs. mic cai. org /mi cca i-20 25/ pap er/ 326 7_p ape r.pd f]
RL VQA Validation authors	2 0 2 3	A Reinforcement Learning Approach for VQA Validation	https://www.sciencedirect.com/science/article/pii/S136184152300083X	Uses RL to validate and improve generic VQA answers; not focused on medical images.	Shows RL can select or filter better answers, supporting NutriDermAI's use of RL to reduce hallucinations and improve reliability.	RL VQ A Vali dati on aut hor s (20 23) . A Rei nfo rce me

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MedThink authors	2 0 2 5	MedThink: A Rationale-Guide d Framework for Explaining Medical Visual Question Answering	https://aclanthology.org/2025.findings-naacl.415.pdf	Creates rationale-genera ting Med-VQA models; not dermatology-spe cific or ABCDE-aware.	Demonstrates how to generate textual rationales alongside answers, aligning with NutriDermAI's goal of JSON explanations and Me dTh ink aut hor s (20 25) . Me dTh

					interpretable responses.	ink: A Ration ale-Gui ded Fra me wor k for Exp lain ing Me dic al Vis ual Qu esti on Ans wer ing [htt ps: //a cla nth olo gy. org /20 25. find ing s-n aac l.41 5.p df]
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HEALMQA authors	2024	HEALMQA: A Healthcare Multimodal Question Answering Benchmark	https://openreview.net/pdf/1d0a6992d0bc72959294fe86ec6553f6cbd3ee7a.pdf	Provides a broad healthcare multimodal QA benchmark; not dermatology or ABCDE-focused.	Offers benchmark design and evaluation ideas for multimodal QA, useful to structure NutriDermAI's test scenarios.	HEALMQA authors (2024). HEALMQA: A Healthcare Multimodal Question Answering Benchmark [https://openreview.net/pdf/1d0a6992d0bc7
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					295 929 4fe 86e c65 53f 6cb d3e e7a .pd f]
Lozano et al.	2 0 2 5	Synthesizing High-Quality Visual Question Answering from Medical Literature	https://arxiv.org/ html/2510.2586 7	Generates VQA samples from text; not focused on dermatology images or ABCDE.	Shows how to synthesize medically grounded VQA data, which can be adapted to generate extra dermatology QA pairs for training. Loz ano et al. (20 25) . Syn the sizi ng Hig h-Q uali ty Vis ual Qu esti on Ans wer ing fro m Me dic al Lite rat ure [htt ps: //ar xiv.

						org /ht ml/ 251 0.2 586 7]
Sviridov et al.	2 0 2 5	Medical Multimodal Multi-agent Dialogue Benchmark	https://aclanthology.org/2025.emnlp-main.1353.pdf	Designs a multi-agent medical dialogue benchmark; not specific to dermatology or ABCDE rules.	Highlights multi-agent setups and metrics for diagnostic dialogue, offering patterns for NutriDermAI's dialogue and escalation design.	Sviridov et al. (2025). Medical Multimodal Multi-agent Dialogue Benchmark [https://aclanthology.org/2025.emnlp-main.1

						353 .pdf]
Dermatology Chatbot authors	2024	Dermatology Chatbot: An AI-Driven Solution for Accessible Skin Care	https://www.ama-assn.org/publications/jama/article/3773	Implements an image+text dermatology chatbot with Hybrid U-Net and MobileNet-V3 but without multimodal LLMs, ABCDE rules, or RL-based VQA.	Shows real-world implementation of dermatology chatbot integrating CNNs and rule-based logic, validating NutriDermAI's multi-stage assistant concept.	Der ma tol o gy Ch at b o t a u t h o r s (2024) . Der ma tol o gy Ch at b o t: An AI- Dri ven Sol utio n for Acc ess ible Ski n Car e [htt ps: //w ww. am eric asp g.c

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Case study authors	2 0 2 4	Design Considerations from a Wizard-of-Oz Dermatology Case Study with a Conversational Agent	https://dl.acm.org/doi/10.1145/3613905.3651891	Explores semi-human-in-t he-loop dermatology conversational interface; not fully automated or multi-stage like NutriDermAI.	Identifies UX and safety considerations (e.g., disclaimers, escalation) for dermatology chatbots, guiding frontend and interaction design. Cas e stu dy aut hor s (20 24) . Des ign Co nsi der atio ns fro m a Wiz ard-of- Oz Der ma tolo gy Cas e Stu dy wit h a Co nve rsa tion al Ag

					ent [htt ps: //dl .ac m.o rg/ doi /10 .11 45/ 361 390 5.3 651 891]
AMIE authors	2 0 2 5	Towards Conversational Diagnostic Artificial Intelligence (AMIE)	https://www.nature.com/articles/s41586-025-08866-7	Provides a general-purpose conversational diagnostic LLM; not dermatology- or ABCDE-specific.	Gives evaluation framework comparing AI vs clinicians and safety practices, which NutriDermAI can adapt for validation and deployment.

					[https://www.nature.com/articles/s41586-025-08866-7]
MIRA framework authors	2025	A Novel Framework for Fusing Modalities in Medical Retrieval-Augmented Generation	https://arxiv.org/html/2507.07902v1	General medical RAG fusion framework; not specifically dermatology or ABCDE-oriented.	Introduces attention-based modality fusion with query-dependent weighting, informing NutriDermAI's IT Fusion and retrieval design.
					MIRA framework authors (2025). A Novel Framework for Fusing Modalities in Medical Ret

						reviewal-Augmented Generation [https://arxiv.org/html/2507.07902v1]
Transformer Attention Fusion authors	2025	Transformer Attention Fusion for Fine-Grained Medical Image Analysis	https://pmc.ncbi.nlm.nih.gov/articles/PMC12216456/	Applies dual attention mechanisms to other medical images (e.g., retinopathy); not dermatology-focused.	Shows that combining self- and cross-attention at multiple scales improves fine-grained recognition, relevant for detailed lesion representation.	Transformer Attention Fusion authors (2025). Transformer Attention

					Fusion for Fine-Gained Medical Image Analysis [https://pubmed.ncbi.nlm.nih.gov/articles/PMC12216456/]
Attention Mechanism review authors	2023	Deep Learning Attention Mechanism in Medical Image Analysis	https://www.scitep.com/journals/ijndi/articles/2504000011	Review of attention mechanisms across medical imaging tasks; no specific dermatology pipeline.	Synthesizes attention patterns that can be plugged into SLS, HC, and IT Fusion modules for NutriDermAI.
					Attention Mechanism review authors (2023)

					. Deep Learning Attention Mechanism in Medical Image Analysis [https://www.sciltp.com/journals/ijndi/articles/25040010011]
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Swin Medical authors	2025	A Multi-Scale Attention-Based Swin Transformer Model for Medical Image Segmentation	https://www.nature.com/articles/s41598-025-22649-0	Generic Swin-based segmentation model; not dermatology-specific.	Validates Swin Transformers with multi-scale attention for segmentation, reinforcing their use in NutriDermAI's SLS.	Swin Medical authors (2025). A Multi-Scale Attention-Based Swin Transformer Model for Medical Image Segmentation [https://www.nature.co
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ISIC organizers	2 0 1 6 - 2 0 2 4	ISIC 2016–2024 Skin Lesion Segmentation and Classification Datasets	https://challenge.isic-archive.com/	Provide benchmark segmentation and classification datasets, but no multi-stage conversational system.	Supply core training and evaluation data for SLS and HC stages (including ISIC 2024 SLICE-3D extension) consistent with NutriDermAI's dataset plan.	ISI C org ani zer s (20 16- 202 4). ISI C 201 6– 202 4 Ski n Les ion Seg me nta tion and Cla ssif icat ion Dat ase ts [htt ps: //c hall

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Tschandl et al.	2018	HAM10000: A Large Collection of Multi-Source Dermatoscopic Images	https://www.nature.com/articles/sdata2018161	Dermatoscopic dataset; does not include pipeline or multimodal reasoning.	Offers 10,015 labeled images across seven lesion types, supporting HC training and evaluation.	Tschandl et al. (2018). HAM10000: A Large Collection of Multi-Source Dermatoscopic Images [https://www.nature.com/article

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Combalia et al.	2019	BCN20000: Dermoscopic Lesions in the Wild	https://www.nature.com/articles/s41597-024-03387-w	Clinical and dermoscopy dataset; not coupled with VQA or ABCDE logic.	Provides large-scale clinical images that extend HC to real-world clinical photos beyond dermoscopy.	Co mb alia et al. (20 19) . BC N2 000 0: Der mo sco pic Les ion s in the Wil d [htt ps: //w ww. nat ure. co m/ arti cle s/s 415 97- 024 -03 387 -w]

He et al.	2020	PathVQA: 30000+ Questions for Medical Visual Question Answering	https://arxiv.org/abs/2003.10286	Pathology VQA dataset; not dermatology-specific.	Pioneers medical VQA dataset design and evaluation metrics transferable to dermatology VQA design.	He et al. (2020). PathVQA: 30000+ Questions for Medical Visual Question Answering [https://arxiv.org/abs/2003.10286]
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Rajkhowa et al.	2024	TM-PathVQA: 90000+ Textless Multilingual Questions for Pathology VQA	https://www.isc-a-archive.org/interspeech_2024/rajkhowa24_interspeech.pdf	Extends PathVQA to multilingual settings; not focused on skin.	Shows that multi-lingual medical VQA is feasible, suggesting multilingual goals for NutriDermAI.	Rajkhowa et al. (2024). TM-PathVQA: 90000+ Textless Multilingual Questions for Pathology VQA [https://www.isc-a-archive.org/interspeech_2024/rajk
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Yim et al.	2024	DermaVQA: Multilingual Visual Question Answering Dataset for Dermatology	https://papers.miccai.org/miccai-2024/211-Paper2444.html	Dermatology VQA dataset; does not define a full assistant system.	Directly supplies dermatology VQA data for training and evaluating NutriDermAI's VQA module.	Yim et al. (2024). DermaVQA: Multilingual Visual Question Answering Dataset for Dermatology [https://papers.miccai.org

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Johnson et al.	2016	MIMIC-III Clinical Database v1.4	https://physionet.org/content/mimiciii/1.4/	ICU clinical text dataset; not dermatology-focused.	Provides large-scale clinical notes for pretraining MedProc's NLP models and reasoning components.	Joh nso n et al. (20 16) . MI MI C-II I Clin ical Dat aba se v1. 4 [htt ps: //p hys ion et.o rg/ con ten t/m imi ciii/ 1.4 /]

Hyland et al.	2023	MIMIC-IV-Note: Deidentified Free-Text Clinical Notes	https://physionet.org/content/mimic-iv-note/	Updated ICU text dataset; not specific to skin diseases.	Offers more recent clinical notes to improve general medical reasoning in MedProc.	Hyland et al. (2023). MIMIC-IV-Note: Deidentified Free-Text Clinical Notes [https://physionet.org/content/mimic-iv-note/]
Lau et al.	2018	VQA-RAD: Radiology Visual Question Answering Dataset	https://collab.dvb.bayern/spaces/TUMdlma/pages/73379920/	Radiology VQA dataset rather than dermatology.	Provides early Med-VQA design and question taxonomy useful for structuring dermatology VQA.	Lau et al. (2018). VQA-RAD:

						Rad iolo gy Vis ual Qu esti on Ans wer ing Dat ase t [htt ps: //c olla b.d vb. bay ern /sp ace s/T UM dlm a/p age s/7 337 992 0/]
Lin et al.	2 0 2 3	Medical Visual Question Answering: A Survey	https://www.sciencedirect.com/science/article/abs/pii/S0933365723001252	Survey of Med-VQA; no specific new model.	Summarizes methods and datasets for Med-VQA, informing NutriDermAI's VQA design space.	Lin et al. (20 23) . Me dic al Vis ual Qu esti

					on Ans wer ing: A Sur vey [htt ps: //w ww. sci enc edir ect. co m/ sci enc e/a rticl e/a bs/ pii/ S09 333 657 230 012 52]
Natural Language Processing of MIMIC authors	2 0 1 9	Natural Language Processing of MIMIC-III Clinical Notes	https://arxiv.org/ pdf/1912.12397. pdf	Applies NLP to clinical notes; not dermatology-spe cific.	Shows how to preprocess and model clinical free-text for entity and concept extraction, informing MedProc's pipeline. Nat ural Lan gua ge Pro ces sin g of MI MI C aut hor s

					(2019). Natural Language Processing of MICLIC-III Clinical Notes [https://arxiv.org/pdf/1912.12397.pdf]
Clinical Text Review authors	2020	Clinical Text Data in Machine Learning: Systematic Review	https://pmc.ncbi.nlm.nih.gov/articles/PMC7157505/	Systematic review; no new model.	Outlines best practices and challenges for clinical NLP, guiding MedProc design. Clinical Text Review authors (2020). Clin

					ical Tex t Dat a in Ma chi ne Lea rnin g: Sys te ma tic Rev iew [htt ps: //p mc. ncb i.nl m.n ih.g ov/ arti cle s/P MC 715 750 5/]
ChatGPT Dermatology review authors	2 0 2 5	The Role of ChatGPT in Dermatology Diagnostics	https://pmc.ncbi.nlm.nih.gov/articles/PMC12191462/	Review of LLM usage; not a specific deployed system like NutriDermAI.	Highlights strengths, risks, and limitations of LLMs in dermatology, supporting safe integration strategies. Ch atG PT Der ma tolo gy revi ew aut hor s (20

					25) . The Role of ChatGPT in Dermatology Diagnostics [https://pmc.ncbi.nlm.nih.gov/articles/PMC12191462/]
Conversational Agents Effectiveness authors	2025	Effectiveness of AI-Driven Conversational Agents in Improving Dermatological Patient Engagement	https://www.jmir.org/2025/1/e69639/	Evaluates chat agents; does not propose NutriDermAI-like pipeline.	Provides evidence that conversational agents improve patient engagement, supporting NutriDermAI's chat interface. Conversational Agents Effectiveness

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Medical Multimodal LLM review authors	2024	Medical Multimodal Large Language Models: A Systematic Review	https://www.sciencedirect.com/science/article/pii/S2950261625000512	Review of MMLLMs; not dermatology-specific.	Summarizes current multimodal LLM architectures and limitations, helping position NutriDermAI relative to state of the art.	Medical Multimodal LLM review authors (2024). Medical Multimodal Large Language Models: A Systematic Review [https://www.scienc

					edir ect. co m/ sci enc e/a rticl e/p ii/S 295 026 162 500 051 2]
MedVLThinker authors	2 0 2 5	MedVLThinker: Simple Baselines for Multimodal Medical Reasoning	https://arxiv.org/ html/2508.0266 9v2	Proposes simple baselines for multimodal reasoning; not dermatology-foc used.	Shows that carefully designed simple multimodal baselines can perform competitively, suggesting efficient designs suited to 12GB hardware.
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Validation Dermatology MLLM authors	2 0 2 5	Validation of a Dermatology-Fo cused Multimodal Large Language Model	https://pmc.ncbi.nlm.nih.gov/articles/PMC12608998/	Validates a dermatology MLLM clinically; does not describe a full multi-stage assistant pipeline.	Demonstrates clinical benefit of domain-specific multimodal models, supporting NutriDermAI's specialization strategy. Vali dati on Der ma tolo gy ML LM aut hor s (20 25) . Vali dati on of a Der ma tolo gy- Foc use d Mul tim oda l Lar ge

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ImageCLEF MEDIQA-MAGI C organizers	2 0 2 5	ImageCLEFmed MEDIQA-MAGIC: Multimodal Medical Dialogue and Generation Benchmark	https://www.imageclef.org/2025/medical/mediqua	Benchmark challenge; not a specific assistant.	Provides task definitions and metrics for multimodal medical dialogue and QA that NutriDermAI can use for evaluation.	Ima ge CL EF ME DIQ A- MA GIC org ani zer s (20 25) . Ima ge CL EF me d ME DIQ

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WoundcareVQA authors	2025	WoundcareVQA: A Multilingual Visual Question Answering Dataset for Wound Care	https://www.sciencedirect.com/science/article/pii/S153204625001170	Wound care VQA dataset; adjacent but not dermatology-specific.	Illustrates the construction of skin-related multilingual VQA datasets, providing patterns transferable to dermatology.	WoundcareVQA authors (2025). WoundcareVQA: A Multilingual Visual Question Answering Dataset for Wound Care [https://www.sciencedirect.co
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Skin Cancer Classification Perspectives authors	2 0 2 4	Decoding Skin Cancer Classification: Perspectives, Insights, and Future Directions	https://www.nature.com/articles/s41598-024-81961-3	Perspectives review; not an implementation of a pipeline.	Provides technical and clinical perspectives on skin cancer AI, informing HC design and evaluation choices. Ski n Ca nce r Cla ssif icat ion Per spe ctiv es aut hor s (20 24) . Dec odi ng Ski n Ca nce r Cla ssif icat ion: Per spe

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